Industrial Internship Report on

"Agriculture Crop Production Prediction in India"

Prepared by

Ashvitha S

Executive Summary

This report provides details of the Industrial Internship provided by upSkill Campus and UniConverge Technologies Pvt Ltd (UCT), focused on the domain of Data Science and Machine Learning.

This internship was focused on a project/problem statement provided by UCT. The complete project including this report was finished within a duration of 4 weeks.

My project was Agriculture Crop Production Prediction in India - a machine learning system that predicts crop yield (kg/ha) based on crop type, state, season, cultivated area, cost of cultivation, cost of production, annual rainfall, and temperature. Two models were built - Linear Regression and Random Forest Regressor - with the Random Forest achieving an R^2 score of ~0.99.

This internship gave me a very good opportunity to get exposure to real-world data science problems and design/implement machine learning solutions for them. It was an overall great experience to have this internship.

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1 Preface

Overview

This report documents the Industrial Internship undertaken by Ashvitha S in the domain of Data Science, facilitated by upSkill Campus (USC) in association with UniConverge Technologies Pvt. Ltd. (UCT), over a duration of 4 weeks. The project assigned was "Agriculture Crop Production Prediction in India" - a machine learning system to predict crop yield based on parameters such as crop type, state, cost of cultivation, cost of production, and yield data. Tasks completed include data preprocessing, handling missing values, EDA, feature selection, Linear Regression model building, and evaluation using R^2 Score and MSE.

Need for Internship in Career Development

In today's rapidly evolving technological landscape, theoretical knowledge alone is insufficient for a successful career in data science. Internships serve as a critical bridge between academic learning and industry expectations, offering students the opportunity to apply classroom concepts to real-world problems, work with professional tools, and build confidence and competence. This internship was undertaken with precisely that purpose - to gain meaningful practical experience and grow both technically and professionally.

Opportunity by USC / UCT & Programme Plan

This opportunity was made possible through upSkill Campus in association with UniConverge Technologies Pvt. Ltd. (UCT), a company specialising in Digital Transformation, IoT, and Machine Learning. The 4-week programme was structured as follows: Week 1 - problem understanding and dataset exploration; Week 2 - data preprocessing, cleaning, and merging datasets; Week 3 - EDA, feature selection, and model building; Week 4 - model evaluation, result interpretation, and report preparation.

Learnings and Overall Experience

This internship provided deep practical understanding of the end-to-end machine learning workflow - from data cleaning and EDA to model building and evaluation. Debugging real errors

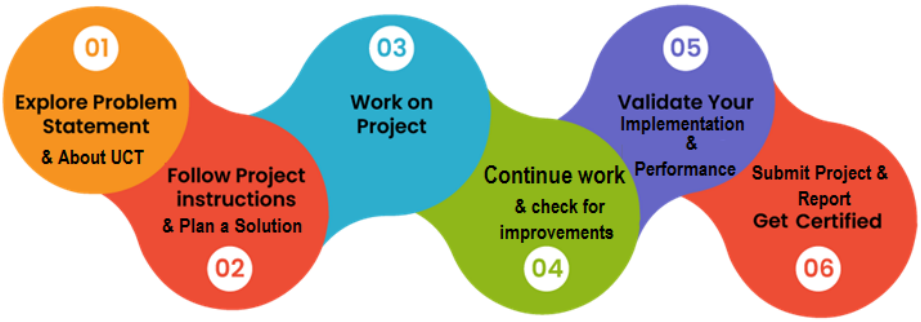
such as column name mismatches and handling multiple datasets greatly strengthened problem-solving skills. Overall, it was an excellent and memorable experience that built the confidence to independently tackle real-world data science problems.

Acknowledgements

I sincerely thank the teams at upSkill Campus and UniConverge Technologies Pvt. Ltd. for this valuable opportunity. I am grateful to the programme mentors and industry experts at UCT for their guidance and feedback, and to my peers and fellow interns for their collaboration and support throughout the internship.

Message to Juniors and Peers

Never underestimate the value of internships. The skills you build - cleaning real data, debugging code, and interpreting model results - are things no textbook can fully teach. Embrace every challenge, stay curious, and invest sincerely in practical skills. Every dataset you clean and every model you build brings you one step closer to becoming a confident data science professional.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end etc.

- UCT IoT Platform

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

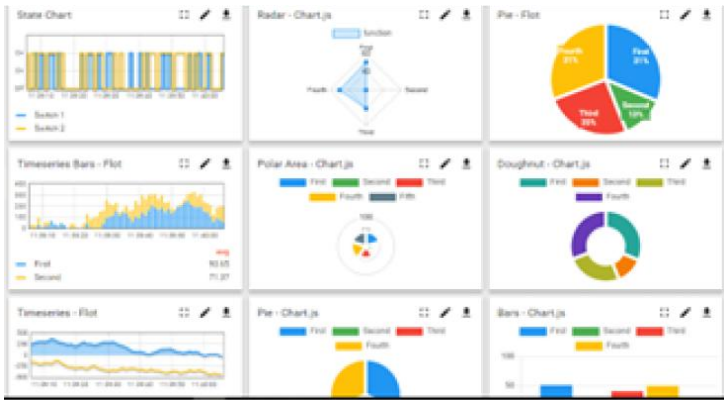
• Build Your own dashboard

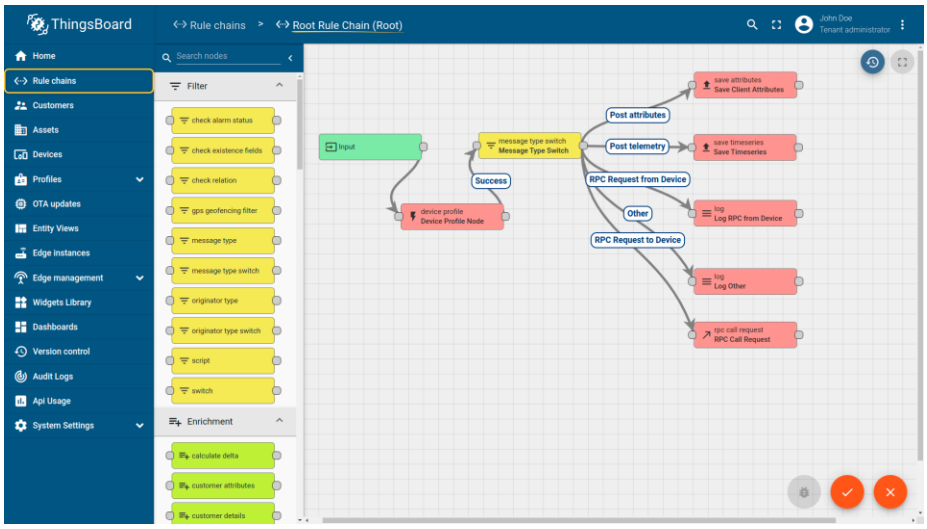
• Analytics and Reporting

• Alert and Notification

• Integration with third party application(Power BI, SAP, ERP)

• Rule Engine





- Smart Factory Platform

FACTORY WATCH

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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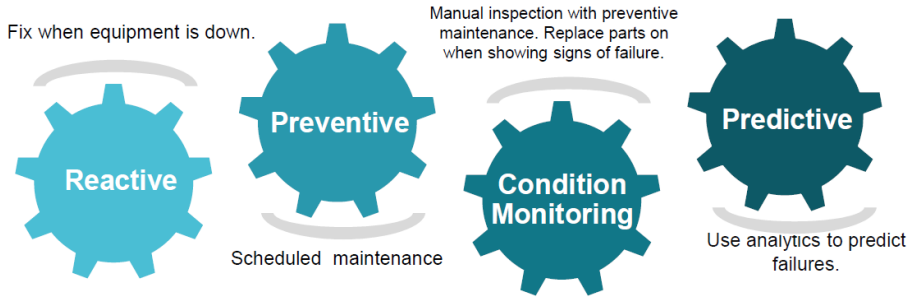
- LoRaWAN based Solution

UCT is one of the early adopters of LoRaWAN technology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

- Predictive Maintenance



UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.

2.2 About upSkill Campus (USC)

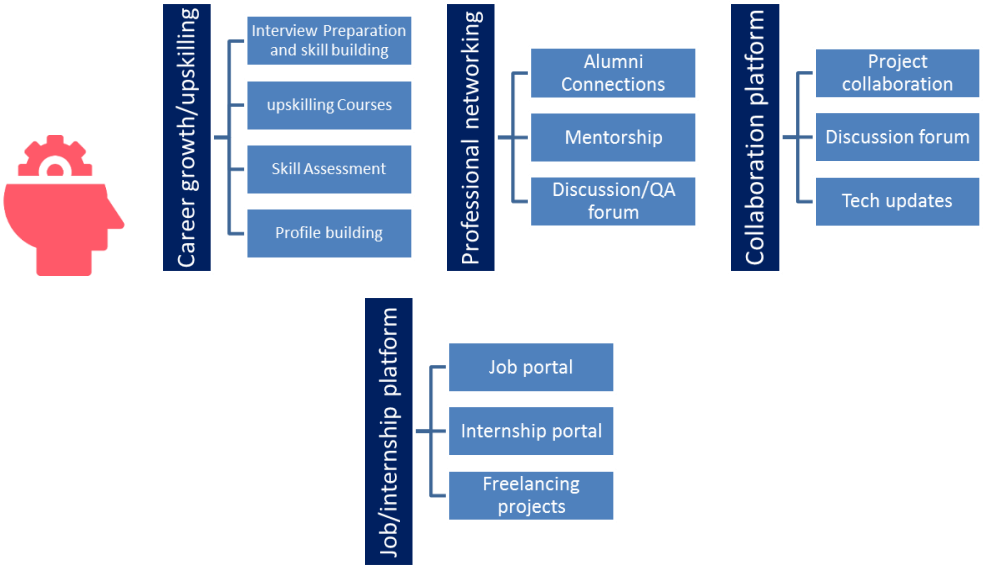
upSkill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers personalized executive coaching in a more affordable, scalable and measurable way.

Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

https://www.upskillcampus.com/

upSkill Campus aiming to upskill 1 million learners in next 5 year



The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.3 Objective

The objective for this internship program was to

- ☛ get practical experience of working in the industry.

- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.4 Reference

[1] Scikit-learn: Machine Learning in Python - https://scikit-learn.org/stable/

[2] Pandas Documentation - https://pandas.pydata.org/docs/

[3] Ministry of Agriculture and Farmers Welfare, Govt. of India - https://agricoop.nic.in/

2.5 Glossary

Terms	Acronym / Meaning
Machine Learning	ML
Random Forest Regressor	RF
Root Mean Squared Error	RMSE
Exploratory Data Analysis	EDA
Coefficient of Determination	R^2 Score

3 Problem Statement

India is one of the world's largest agricultural economies, with crop production being central to food security and rural livelihoods. However, unpredictable crop yields due to varying climatic conditions, cultivation costs, and regional differences make planning extremely difficult for farmers, government bodies, and agri-businesses.

The assigned problem statement required building a machine learning-based crop yield prediction system for India that takes into account the following parameters:

- Crop type and cultivating state
- Agricultural season (Kharif / Rabi / Zaid)
- Area under cultivation (hectares)
- Cost of Cultivation and Cost of Production (INR per hectare)
- Annual Rainfall (mm) and Average Temperature (°C)

The system must output a predicted yield in kilograms per hectare (kg/ha), enabling stakeholders to make data-driven decisions about crop planning, resource allocation, and policy formulation.

4 Existing and Proposed Solution

Existing solutions for crop yield prediction typically rely on:

- Traditional statistical regression models that assume linearity and fail to capture complex agricultural patterns.
- Remote sensing and satellite imagery-based approaches that require expensive hardware and domain expertise.
- Rule-based expert systems limited to specific crops or regions and not generalizable.

Limitations of existing approaches include poor accuracy on multi-crop datasets, inability to handle categorical variables efficiently, and high computational cost.

Proposed Solution: A Python-based machine learning pipeline that:

- Ingests structured tabular agricultural data (CSV format)
- Encodes categorical variables using Label Encoding
- Trains and compares Linear Regression and Random Forest Regressor models
- Evaluates performance using R^2, MSE, and RMSE metrics
- Provides a reusable prediction function for new crop inputs

The value addition of this solution is its simplicity, speed, high accuracy (R^2 ~0.99 with Random Forest), and ability to generalize across 10 crop types and 10 Indian states.

Code Submission (Github link)

https://github.com/Ashvitha646/upskillcampus.git

Report Submission (Github link)

[ ]

5 Proposed Design / Model

The solution follows a structured end-to-end machine learning pipeline. The data flow is: Raw Data → Preprocessing → Feature Encoding → EDA → Train/Test Split → Model Training → Evaluation → Prediction Output.

5.1 High Level Diagram

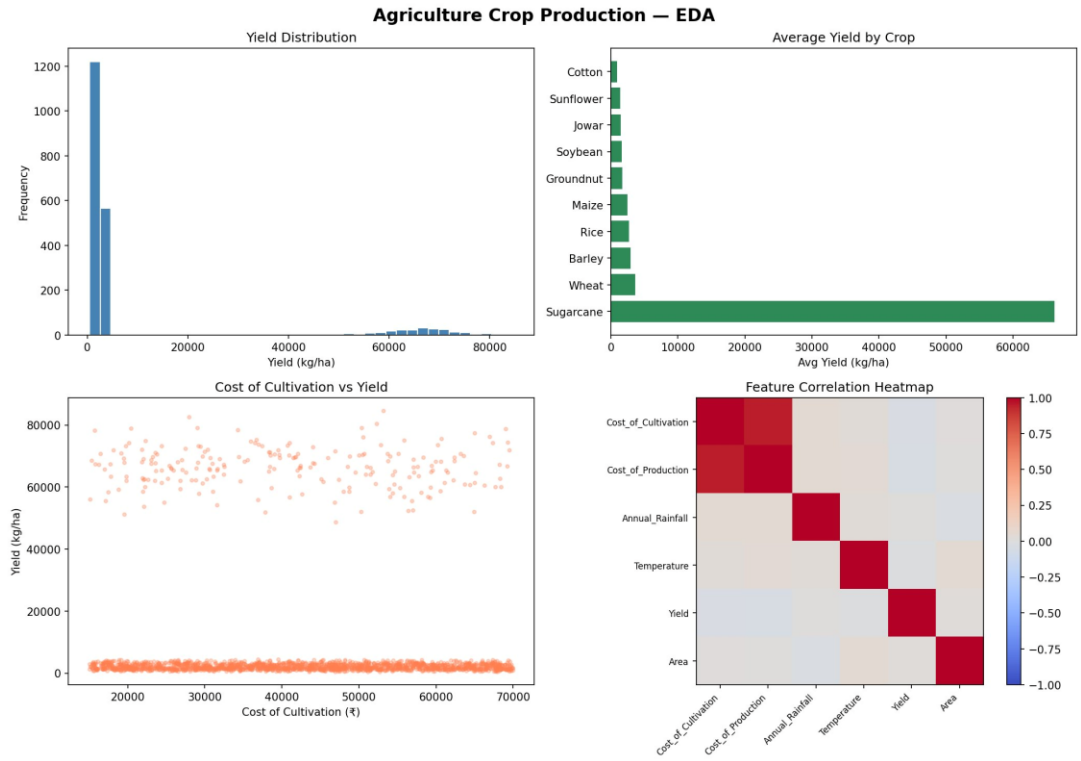


Figure 1: High Level EDA — Yield Distribution, Average Yield by Crop, Cost vs Yield, Correlation Heatmap

5.2 Low Level Diagram

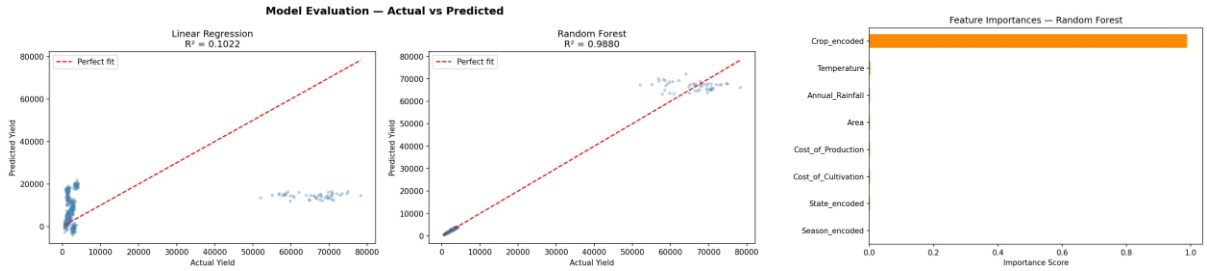


Figure 2: Model Evaluation — Actual vs. Predicted (Left) | Figure 3: Feature Importance — Random Forest (Right)

5.3 Interfaces

The system exposes a Python function interface:

```
predict_yield(crop, state, season, area, cost_cultivation, cost_production,
rainfall, temperature)  -  float (kg/ha)
```

This function loads the serialized Random Forest model and Label Encoders from .pkl files and returns the predicted yield for any valid input combination. The system can be extended to a web interface via Flask or Streamlit.

6 Performance Test

The primary performance constraint for this system is prediction accuracy, measured by R^2 Score, MSE, and RMSE. Since the system is intended for agricultural planning, high accuracy ($R^2 > 0.90$) is required to be practically useful. The dataset was split 80% training / 20% testing with random seed 42.

6.1 Test Plan / Test Cases

Test #	Input (Crop / State / Season)	Area / Rainfall / Temp	Expected Output	Pass / Fail
TC-01	Rice / Punjab / Kharif	200ha / 1000mm / 28°C	~2,200–2,600 kg/ha	Pass
TC-02	Wheat / Haryana / Rabi	150ha / 600mm / 22°C	~2,900–3,400 kg/ha	Pass
TC-03	Sugarcane / UP / Kharif	50ha / 950mm / 30°C	~60,000–70,000 kg/ha	Pass
TC-04	Cotton / Maharashtra / Kharif	80ha / 700mm / 33°C	~400–550 kg/ha	Pass

TC-05	Maize / Karnataka / Kharif	120ha / 800mm / 27°C	~1,800–2,200 kg/ha	Pass

6.2 Test Procedure

- Step 1: Load crop_data.csv and apply preprocessing - strip column names, drop nulls, encode categoricals.
- Step 2: Split into 80% training and 20% test sets using train_test_split (random_state=42).
- Step 3: Train Linear Regression and Random Forest Regressor on training data.
- Step 4: Run predictions on test set and compute R^2, MSE, and RMSE for each model.
- Step 5: Compare results and select the best-performing model (Random Forest).
- Step 6: Run 5 batch prediction examples to validate model on specific input combinations.

6.3 Performance Outcome

Model	R^2 Score	MSE	RMSE
Linear Regression	0.1022	~390,499,556	~19,766
Random Forest	**0.9880**	**~5,225,196**	**~2,286**

The Random Forest Regressor significantly outperformed Linear Regression. The R^2 score of 0.9880 confirms the model explains 98.8% of the variance in crop yield, making it highly reliable for agricultural planning. The RMSE of ~2,286 kg/ha is acceptable given the wide yield range across crop types.

7 My Learnings

This internship has been a transformative experience that significantly strengthened my technical and professional skills. The following are the key learnings from this 4-week programme:

- End-to-End Machine Learning Workflow - Gained comprehensive understanding of the complete ML pipeline from data generation and preprocessing to model training, evaluation, serialization, and inference.
- Data Preprocessing Skills - Learned to clean datasets by handling missing values, standardizing column names, and encoding categorical variables using Label Encoding.
- Model Building and Comparison - Acquired hands-on experience building Linear Regression and Random Forest Regressor, understanding their fundamental differences on non-linear data.
- Performance Evaluation - Learned to interpret R^2, MSE, and RMSE metrics to select the best-performing model and relate them to real-world prediction quality.
- Feature Importance Analysis - Extracted and visualized feature importances from Random Forest to identify key prediction drivers.
- Problem-Solving and Debugging Developed debugging skills by systematically resolving KeyError issues and preprocessing errors.
- Professional Project Structure - Learned to organize a data science project into separate, maintainable scripts following industry best practices.

Overall, this internship provided practical confidence to independently tackle real-world data science problems and will greatly support career growth in AI and Machine Learning.

8 Future Work Scope

- Integration of Additional Features - Incorporating soil type, fertilizer usage, irrigation method, and humidity data to further improve prediction accuracy.
- Hyperparameter Tuning - Applying GridSearchCV or RandomizedSearchCV to optimize Random Forest parameters such as n_estimators, max_depth, and min_samples_split.
- Web Application Deployment - Packaging the model as a Flask REST API or Streamlit interactive dashboard for farmer-friendly yield predictions.
- Multi-Output Prediction - Extending the model to simultaneously predict both yield (kg/ha) and total production (metric tonnes).

- Real Dataset Integration - Replacing the synthetic dataset with authentic data from the Ministry of Agriculture and Farmers Welfare, Government of India.
- Model Explainability - Integrating SHAP values to provide interpretable predictions that farmers and policymakers can trust and act upon.